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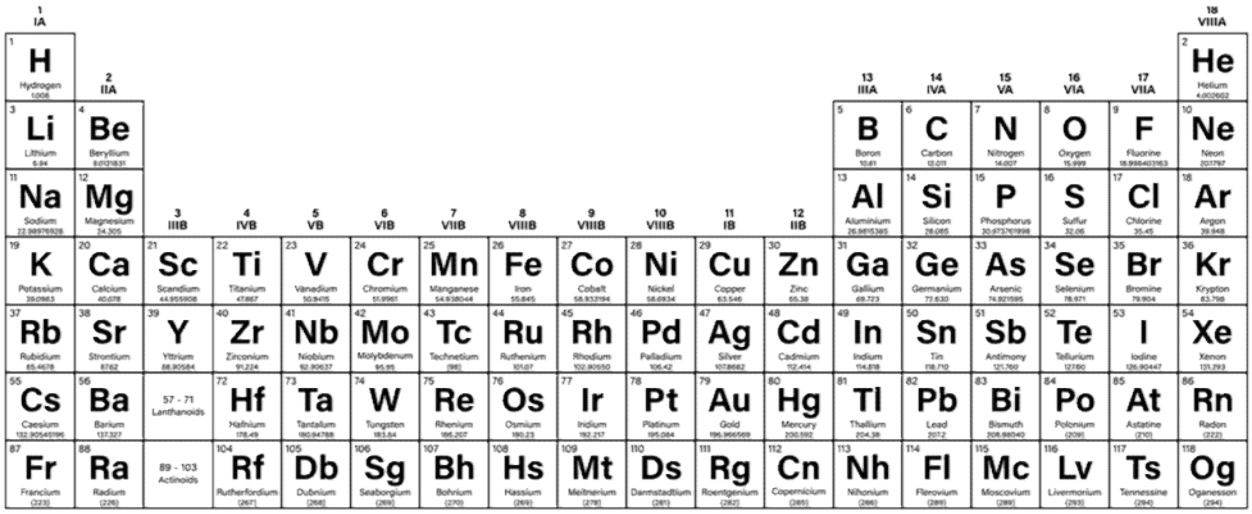
Instructions for CM2021 Mid-term Test

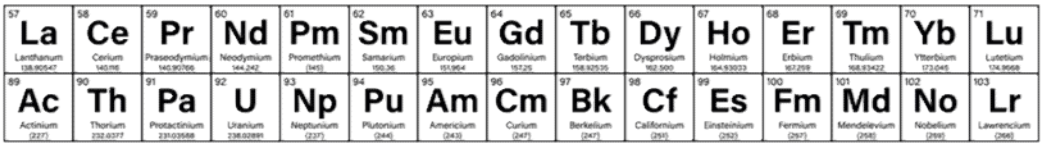
1. This test paper contains twenty (20) multiple-choice questions

2. Answer ALL questions (1 mark each)

3. Time allowed: 1 hour

4. Do not turn over the test paper until told to do so





1) Which of the coordination compounds will give exactly three isomers?

a. $[Fe(en)_3]Cl_3$ b. $[Co(en)_2Cl_2]Cl$

c. $[Fe(PPh_3)_3(NH_3)ClBr]Cl$ d. $[Co(PPh_3)_3Cl]Cl_3$

2) Hybridization involves:

a. addition of an electron pair b. combination and redistribution of atomic orbitals

c. removal of an electron pair d. separation of orbitals

3) The name of $[Ni(NH_3)_4][NiCl_4]$ is:

a. tetraamminenickel(II) tetrachloronickelate(II)

b. tetraamminenickel(II) tetrachloronickel(II)

c. tetraamminenickel(0) tetrachloronickelate(0)

b. tetraamminenickel(0) tetrachloronickel(0)

4) The name of $K_3[Ir(C_2O_4)_3]$ is:

a. potassium trioxalatoiridium(III) b. potassium trioxalatoiridate(III)

c. potassium tris(oxalato)iridium(III) d. potassium tris(oxalato)iridate(III)

5) Which of the following octahedral complexes will have the highest Δo?

a. $[Co(H_2O)_6]^{3+}$ b. $[Co(NH_3)_6]^{3+}$

c. $[Co(CN)_6]^{3-}$ d. $[CoCl_6]^{3-}$

6) How many isomer(s) is/are possible for a tetrahedral complex with the formula $[M(A\text{-}A)_2]$? Note that A-A refers to a symmetrical bidentate ligand.

a. 1 b. 2

c. 3 d. 4

7) Which of the following has the highest molar conductivity?

a. $[Co(NH_3)_6]Cl_3$ b. $[Co(NH_3)_5Cl]Cl_2$

c. $[Co(NH_3)_4Cl_2]Cl$ d. $[Co(NH_3)_4BrF]Cl$

8) The ionization isomer of $[Cr(H_2O)_4Cl(NO_2)]Cl$ is:

a. $[Cr(H_2O)_4Cl_2](NO_2)$ b. $[Cr(H_2O)_4Cl(ONO)]Cl$

c. $[Cr(H_2O)_3Cl_2(NO_2)].H_2O$ d. All of the above

9) The C-O bond length of CO ligand increases as we change the central metal ion from a d^0 species to a d^1 species. Assuming that the ion radii and oxidation states of both species are the same, the observation is due to:

a. ligand-to-metal π bonding b. metal-to-ligand π-back bonding

c. σ bonding d. none of the above

10) Which of the following statements is true?

a. $E(d_{xy}) > E(d_z^2)$ in both tetrahedral and octahedral complexes

b. $E(d_{xy}) < E(d_z^2)$ in both square planar and octahedral complexes

c. $E(d_{xy}) > E(d_z^2)$ in tetrahedral but $E(d_{xy}) < E(d_z^2)$ in octahedral complexes

d. $E(d_{xy}) < E(d_z^2)$ in square planar but $E(d_{xy}) < E(d_z^2)$ in octahedral complexes

11) Cisplatin compounds are used in the treatment of:

a. malaria b. cancer

c. AIDS d. yellow fever

12) Which of the following isomerisms comprise of metal complex isomers with different general molecular formula?

a. optical isomerism b. coordination isomerism

c. fac/mer isomerism d. none of the above

13) $[Cu(SCN)_2(NH_3)_4]$ is an octahedral complex. How many isomers are possible for this molecular formula?

a. 4 b. 6

c. 8 d. 12

14) A dextrorotatory optical isomer rotates plane-polarized light to the _____ and is denoted as _____.

a. left; d b. right, d

c. left, l d. right, l

15) The complex ion $[NiCl_6]^{4-}$ has a crystal field splitting parameter of 81 KJ/mol. Calculate its crystal field stabilization energy (CFFE).

a. 97 KJ/mol
b. 227 KJ/mol
c. 1.2 KJ/mol
d. 65 KJ/mol

16) Ethylenediamine is observed to form a more stable metal complex than methylamine. This is due to:

a. macrocyclic effect
b. lewis basicity
c. pi bonding
d. chelate effect

17) What is the CFSE of low-spin d^7 metal complex?

a. 1.8Δo
b. 0.8Δo
c. 1.8Δo - P
d. 1.8Δo - 3P

18) Which of the following on CO ligand is true?

a. σ-donor, π-acceptor
b. σ-donor, π-donor
c. σ-acceptor, π-acceptor
d. σ-acceptor, π-acceptor

19) Which of the following metal complexes is expected to give the strongest Jahn-Teller effect?

a. low-spin d^4
b.high-spin d^6
c. low-spin d^7
d. d^2

20) Which of the following causes d-orbitals splitting in crystal field theory?

a. size of the metal ion/atom
b. directionality of metal d orbitals
c. spherical field
d. all of the above